

CAMP DAVID, MD, USA

WMO: 722823

Lat: 39.645N

Lon: 77.468W

Elev: 1842

StdP: 13.74

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03749

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.4	9.9	-7.4	4.0	7.3	-3.2	5.0	13.1	26.8	23.5	24.8	25.8	11.6	330	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	86.7	69.4	84.2	68.6	81.9	67.9	73.9	82.5	72.4	80.2	71.2	78.2	5.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	123.4	78.1	70.0	118.3	76.2	68.9	113.5	75.0	38.6	82.4	37.2	80.3	36.1	78.4	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	19.0	16.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	28.6	30.8	40.1	52.8	62.2	68.5	73.0	71.3	64.9	54.5	43.7	34.8
	DBStd	17.73	11.19	10.92	10.90	9.82	7.91	5.65	4.95	4.84	7.03	9.17	8.68	10.00
	HDD50	2399	669	543	343	75	6	0	0	0	0	56	224	483
	HDD65	5446	1129	957	771	387	148	30	3	8	88	347	640	938
	CDD50	3207	5	6	37	159	384	556	713	660	448	195	33	11
	CDD65	781	0	0	0	21	62	137	251	204	86	20	0	0
	CDH74	4245	0	0	1	202	328	697	1517	1018	431	51	0	0
	CDH80	949	0	0	0	70	48	131	389	213	92	6	0	0
Wind	WSAvg	7.9	10.1	10.3	9.6	8.7	6.9	6.3	5.8	5.4	5.8	7.4	8.5	10.0
Precipitation	PrecAvg	44.0	3.3	2.5	4.1	3.7	3.7	4.0	3.9	3.8	4.6	3.4	3.1	3.5
	PrecMax	68.9	8.3	5.9	8.9	7.6	8.4	8.7	9.6	6.0	9.6	7.6	6.7	6.3
	PrecMin	28.6	1.2	0.3	0.5	1.7	0.9	1.3	1.7	1.5	0.0	0.5	0.5	0.6
	PrecStd	9.1	1.8	1.3	1.8	1.5	1.9	2.2	1.9	1.2	2.7	2.0	1.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.1	62.7	71.1	87.9	84.4	87.8	91.0	89.1	86.1	79.5	67.2	64.1
		MCWB	55.0	55.2	54.8	62.5	67.0	70.8	72.0	70.0	70.6	67.6	57.6	57.6
	2%	DB	55.1	56.5	65.7	79.7	80.8	83.4	87.0	84.8	82.8	74.3	62.7	59.5
		MCWB	49.8	48.9	52.7	59.6	65.2	69.0	71.1	67.7	69.2	64.4	53.0	54.8
	5%	DB	50.1	52.0	61.7	73.2	77.6	80.8	84.2	82.2	79.0	70.9	59.4	54.1
		MCWB	45.1	46.2	51.4	56.7	64.0	67.5	69.6	67.7	67.6	62.7	50.5	50.3
	10%	DB	44.4	46.4	57.2	68.0	74.4	78.1	81.5	79.6	75.2	67.9	56.3	48.7
		MCWB	40.3	40.1	49.2	54.7	62.0	66.2	69.3	67.5	66.3	61.3	49.1	44.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	56.9	58.6	64.5	71.7	74.3	75.9	75.7	73.8	70.5	59.8	59.9
		MCDB	58.3	60.0	65.0	80.8	81.6	84.1	85.5	84.4	83.0	74.8	64.8	62.2
	2%	WB	51.2	51.4	56.0	61.2	69.2	71.8	74.2	73.2	71.4	68.3	55.7	55.1
		MCDB	54.2	54.3	62.1	74.6	77.6	80.5	83.1	80.1	77.7	72.0	59.1	58.6
	5%	WB	45.7	45.8	53.6	59.1	67.4	70.5	72.8	71.4	69.8	65.5	53.2	50.8
		MCDB	48.6	51.7	60.4	70.8	74.2	77.9	81.0	77.4	75.1	69.9	57.3	53.5
	10%	WB	40.3	40.4	49.5	56.7	65.3	68.8	71.4	70.0	68.1	62.1	50.7	44.8
		MCDB	44.5	45.9	57.0	66.3	71.0	75.1	78.9	75.8	73.4	66.9	55.3	48.0
Mean Daily Temperature Range	5% DB	MDBR	12.5	13.9	15.7	17.7	15.4	15.1	14.8	14.3	14.1	13.0	12.9	11.6
		MCDBR	16.2	18.4	20.4	22.6	18.3	17.3	16.8	16.5	16.9	15.0	15.5	16.3
		MCWBR	14.7	14.9	13.7	11.4	9.8	9.0	8.0	7.6	8.1	7.9	10.5	14.7
	5% WB	MCDBR	15.6	18.2	19.1	20.7	15.8	15.5	14.8	14.1	14.4	13.5	15.5	16.3
		MCWBR	15.4	16.4	14.4	12.4	9.1	8.8	7.8	7.8	7.4	7.9	13.0	15.4
Clear-Sky Solar Irradiance	taub	0.290	0.303	0.333	0.372	0.406	0.464	0.481	0.476	0.402	0.343	0.304	0.290	
	taud	2.465	2.407	2.406	2.319	2.248	2.115	2.087	2.082	2.268	2.444	2.518	2.498	
	Ebn at Noon	281	292	293	286	277	260	254	252	266	273	274	272	
	Edh at Noon	25	30	34	39	43	49	50	49	39	29	23	23	
All-Sky Solar Radiation	RadAvg	601	883	1173	1510	1690	1873	1856	1649	1345	981	698	523	
	RadStd	51	104	73	125	183	146	108	86	175	111	62	56	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (63 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	+113	

Nomenclature: See separate page